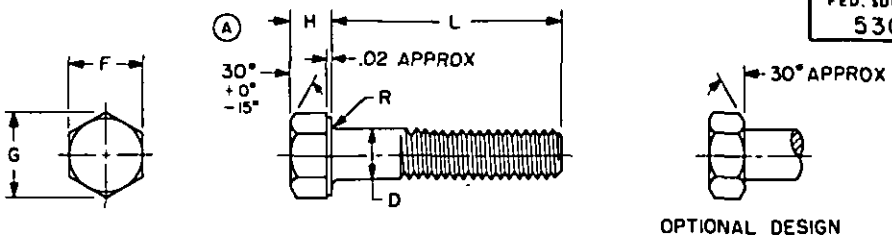
FED. SUP CLASS 5305



REVIEWER: AT, AV, IS, NS
USER: MC, ME, MI, OS, YD

NOMINAL SIZE			.250	.3125	.375	.4375	.500	.5625	.625
THREADS PER INCH (UNC-2A)			20	18	16	14	13	12	11
D BODY DIAMETER	MAX		.250	.3125	.375	.4375	.500	.5625	.625
	MIN		.245	.3065	.369	.4305	.493	.5545	.617
F WIDTH ACROSS FLATS	MAX		.4375	.500	.5625	.625	.750	.8125	.9375
	MIN		.4280	.489	.5510	.612	.736	.7980	.9220
G WIDTH ACROSS CORNERS	MAX		.505	.577	.650	.722	.866	.938	1.083
	MIN		.488	.557	.628	.698	.840	.910	1.051
H HEAD HEIGHT	MAX		.163	.211	.243	.291	.323	.371	.403
	MIN		.150	.195	.226	.272	.302	.348	.378
R RADIUS	MAX	Ⓐ	.025	.025	.025	.025	.025	.045	.045
	MIN	Ⓐ	.015	.015	.015	.015	.015	.020	.020
TENSILE STRENGTH (LBS-MIN)		Ⓐ	2,540	4,190	6,200	8,500	11,350	14,560	18,080

L LENGTH	TOLERANCE Ⓐ SIZE .375 AND UNDER	SIZES .4375 AND .500	SIZES .5625 AND .625	DASH NO.	DASH NO. *	DASH NO.	DASH NO.	DASH NO.	DASH NO. *	DASH NO.
.375	+.000 −.030	+.000 −.030	+.000 −.030	1	27	53				
.438				2*	28	54*	79*			
.500				3	29	55	80	104*		
.562				4*	30	56*	81*	105*	130	
.625				5	31	57	82	106	131	155
.750				6	32	58	83	107	132	156
.875				7	33	59	84	108	133	157
1.000				8	34	60	85	109	134	158
1.125	+.000 −.040	+.000 −.060	+.000 −.080	9*	35	61*	86*	110*	135	159*
1.250				10	36	62	87	111	136	160
1.375				11*	37	63*	88*	112*	137	161*
1.500				12	38	64	89	113	138	162
1.750				13	39	65	90	114	139	163
2.000				14	40	66	91	115	140	164
2.250				15	41	67	92	116	141	165
2.500				16	42	68	93	117	142	166
2.750	+.000 −.060	+.000 −.080	+.000 −.100	17	43	69	94	118	143	167
3.000				18	44	70	95	119	144	168
3.250				19	45	71	96	120	145	169
3.500				20	46	72	97	121	146	170
3.750				21*	47	73*	98*	122*	147	171*
4.000				22	48	74	99	123	148	172
4.250	+.000 −.100	+.000 −.100	+.000 −.100	23*	49	75*	100*	124*	149	173*
4.500				24	50	76	101	125	150	174
4.750				25*	51	77*	102*	126*	151	175*
5.000				26	52	78	103	127	152	176
5.500								128	153	177
6.000								129	154	178

* INACTIVE FOR NEW DESIGN - Dash numbers 27 thru 52, 130 thru 154 and those with an (*) are inactive for new design.

Ⓐ

Ⓐ For changes see sheets 1 and 2.

APPROVED 4 MAR 1954 REVISED Ⓐ 15 FEB 1973

This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. WC
Other Cust SH 82

TITLE: SCREW, CAP, HEXAGON HEAD - NICKEL-COPPER ALLOY, PLAIN FINISH, UNC-2A

MILITARY STANDARD
MS 35311
SHEET 1 OF 2

PROCUREMENT SPECIFICATION: FF-S-85

SUPERSEDES:

DD FORM 1 SEP 67 672-1 (Condensed)

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FED. SUP CLASS 5305

NOMINAL SIZE		.750	.875	1.000	1.125	1.250
THREADS PER INCH (UNC-2A)		10	9	8	7	7
D BODY DIAMETER	MAX	.750	.875	1.000	1.125	1.250
	MIN	.741	.866	.990	1.114	1.239
F WIDTH ACROSS FLATS	MAX	1.125	1.3125	1.500	1.6875	1.875
	MIN	1.100	1.2850	1.469	1.6310	1.812
G WIDTH ACROSS CORNERS	MAX	1.299	1.516	1.732	1.949	2.165
	MIN	1.254	1.465	1.675	1.859	2.066.
H HEAD HEIGHT	MAX	.483	.563	.627	.718	.813
	MIN	.455	.531	.591	.658	.749
R RADIUS	MAX	.045	.065	.095	.095	.095
	MIN	.020	.040	.060	.060	.060
(A) TENSILE STRENGTH (LBS-MIN)		26,720	36,960	48,480	61,040	72,500

L LENGTH	TOLERANCE SIZE .750	TOLERANCE SIZES .875 AND 1.000	TOLERANCE SIZES 1.125 AND 1.250	DASH NO.	DASH NO.	DASH NO.	DASH NO. *	DASH NO.
.750	+.000	+.000	-----	179				
.875	-.030	-.040		180	202			
1.000				181	203	224		
1.125				182*	204*	225*	245	
1.250				183	205	226	246	265
1.375				184*	206*	227*	247	266*
1.500	+.000	+.000	+.000	185	207	228	248	267
1.750	-.080	-.100	-.120	186	208	229	249	268
2.000				187	209	230	250	269
2.250				188	210	231	251	270
2.500				189	211	232	252	271
2.750				190	212	233	253	272
3.000				191	213	234	254	273
3.250	+.000	+.000	+.000	192	214	235	255	274*
3.500	-.100	-.140	-.160	193	215	236	256	275
3.750				194	216	237	257	276*
4.000				195	217	238	258	277
4.250				196*	218*	239*	259	278*
4.500				197	219	240	260	279
4.750	+.000	+.000	+.000	198*	220*	241*	261	280*
5.000	-.100	-.160	-.180	199	221	242	262	281
5.500				200	222	243	263	282
6.000				201	223	244	264	283

(A) * INACTIVE FOR NEW DESIGN - Dash numbers 245 thru 264 and those with an (*) are inactive for new design.

NOTES:

(A) 1. MATERIAL: Nickel-Copper alloy in accordance with QQ-N-281, Class A, having a minimum ultimate tensile strength of 80,000 PSI.
2. PROTECTIVE COATING: None.
3. THREADS: The threads shall be in accordance with Screw-Thread Standards for Federal Services, Handbook H28.
4. DIMENSIONS: All dimensions are in inches, unless otherwise specified.
5. THREAD LENGTH: Minimum thread length shall be twice the diameter plus 0.250 inch. The tolerance shall be plus 0.188 inch or 2-1/2 threads, whichever is greater. On screws that are too short for minimum thread lengths, the distance from the head bearing surface to the first complete thread shall not exceed the length of 2-1/2 threads for sizes thru 1.000 and 3-1/2 threads for sizes over 1.000.
6. PART NUMBER: The MS part number consists of the MS number, plus the dash number. Example: MS35311-1.
7. Referenced documents shall be of the issue in effect on date of invitations for bid.
8. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

REVIEWER: AT, AV, IS, NS (USER: MC, ME, MI, OS, YD

This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

APPROVED 4 MAR 1954 REVISED (A) FOR CHANGES SEE SHEETS 1 AND 2.

P.A. WC	TITLE	MILITARY STANDARD
Other Cust SH 82	SCREW, CAP, HEXAGON HEAD - NICKEL-COPPER ALLOY, PLAIN FINISH, UNC-2A	MS35311
PROCUREMENT SPECIFICATION FF-S-85	SUPERSEDES:	SHEET 2 OF 2

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED.

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